SANAYA VAKHARIA

Machine Learning | Recommender Systems | Product & Data Analytics
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EDUCATION

University of California, San Diego | B.S. Mathematics - Computer Science; Minor: Business - Economics June 2026
Provost Honors | Relevant Coursework: Recommender Systems & Web Mining, Machine Learning, Statistics & Probability, Data Structures & Algorithms, Game Theory

EXPERIENCE

Consultant Intern | Kearney, New York City Jul 2025 - Sep 2025
- Analyzed large datasets in Excel and SQL for a global consulting engagement, translating findings into insights that informed client strategy.
- Automated data workflows and built dashboards to improve data accuracy, efficiency, and quality control across client deliverables.
- Collaborated with international teams and participated in client meetings, communicating analyses across technical and business stakeholders.

Tech Stack: SQL, Excel, Data Analysis, Dashboarding

Machine Learning Engineer | CoCreate x SPYKAR, Remote Jul 2024 - Sep 2024
- Built and deployed a personalized recommendation engine using XGBoost and collaborative filtering, increasing customer repeat visits by 32%.
- Developed a Transformer-based NLP chatbot for dynamic voice and visual recommendations, increasing customer engagement by 41%.
- Fine-tuned CNN and Transformer models for product tagging, improving ad targeting efficiency by 25%.

Tech Stack: XGBoost, Collaborative Filtering, Transformers, CNNs, NLP

Data Analysis Intern | Seagate Technology, Fremont, CA May 2024 - Sep 2024
- Developed ML-based anomaly detection models across 500TB+ of storage performance data, improving failure prediction rates by 23%.
- Integrated Spark, Airflow, and Kafka into real-time processing workflows, reducing ETL latency by 40%.
- Automated Salesforce CRM data processing with NLP-based text classification, streamlining marketing data segmentation by 30%.

Tech Stack: Spark, Airflow, Kafka, NLP, Salesforce CRM

SELECTED PROJECTS

Scene-Aware Product Categorization using ML | Python, XGBoost, Random Forest
- Built an automated product-tagging pipeline using bounding-box feature engineering, achieving 73% classification accuracy.
- Combined latent-factor modeling with scene-based context extraction, improving classification performance by 15%.
- Optimized hyperparameters with GridSearchCV, leading to a 2x increase in model efficiency.

Market Trend Forecasting & Consumer Insights with ML | Python, scikit-learn, NLP, Time-Series, Tableau, SQL
- Built a predictive analytics pipeline using ARIMA, LSTMs, and NLP-based sentiment analysis to identify demand patterns across 500,000+ online product reviews and news articles.
- Forecasted sales and market trends from consumer and news signals, translating model outputs into business-facing insights.
- Developed an interactive Tableau dashboard to track market sentiment and support data-driven business decisions.

TECHNICAL SKILLS

Languages: Python, SQL, Java, R, C++, JavaScript
ML / Data: PyTorch, scikit-learn, XGBoost, Pandas, NumPy, Spark, Airflow, Kafka
Tools: Tableau, Git/GitHub, Jupyter, PostgreSQL, MySQL, Excel, Jenkins

LEADERSHIP

Pro-Dev Events Chair, CSES, UCSD - Organized 15+ AI/ML and career-growth workshops. **Sports Associate Editor, The UCSD Guardian** - Wrote and edited university sports coverage.